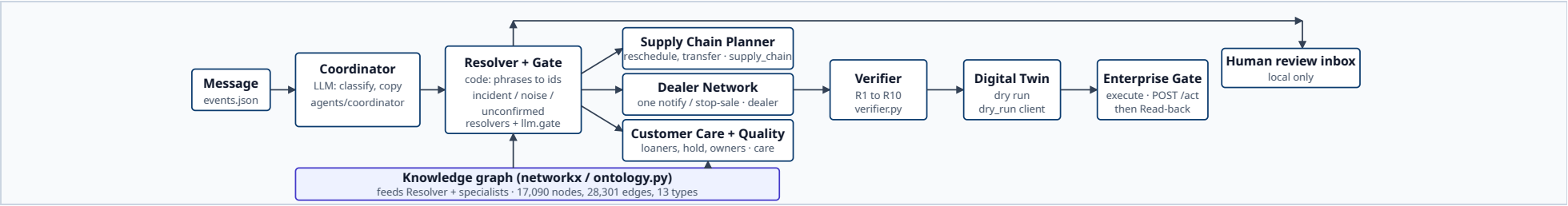
Wacky Wheels Control Tower - Design Card

Hitachi Hack the Highway - TAMU Datathon 2026 - Darshil Rayjada

LLM reads. Code decides. Every action verified.

1. Architecture



2. LLM versus code

LLM	Code
Classify; copy exact phrases; pick a query function in Ask.	Id resolution; relative dates via switches (NEXT_WEEKDAY=following_week TRANSFER_LOWER_BOUND=restart TRANSFER_SEARCH=day_first PLANT_TIEBREAK=plants_csv_order SLOT_PRIORITY=build_date COMPOUND_ORDER=message_order NOTIFY_SCOPE=per_event NO_SLOT_ACTION=escalate ESCALATE_NOISE=False FUZZY_THRESHOLD=90); affected-order search; slot ledger; transfers; loaner ranking; every write.

3. Guardrails

- Grounding check on LLM explanations; rule-based noise cross-check + hard write block on noise.
- Exactly one escalate for unconfirmed; rule 10 affected-only; one dealer.notify per dealer per event.
- Verifier blocks on any failed check; twin dry run; read-back after writes.
- Failed calls: stop remaining plan, escalate with tool + error, no blind retries.
- Locked scored baseline handbook_v1 in Policy Studio (publish makes a new profile only).

5. Explainability and human in the loop

- Per-event summary + reason, with graph path evidence in each trace.
- Human review inbox stores decisions locally; never sent to the enterprise server.

4. Evaluation

Metric	Value	Source file
Events	15 = 11 incidents + 2 escalated + 3 noise	traces/run_summary.json
Writes / reads / audited	766 / 193 / 961	run_summary + self-check
Failed	0	run_summary + self-check
Self-check	15/15 PASS	eval/self_check_report.md
LLM	15 calls; 19,416 tok (17,149 / 2,267)	run_summary.json
Graph	17,090 nodes, 28,301 edges, 13 types	exports/graph_stats.json
Model	qwen2.5:3b-instruct (Ollama, local)	run_summary.json

Checked by independent eval/self_check.py (no control_tower imports) replaying the action log on a fresh enterprise; per-event traces with graph paths; audit trail by rule.

TC map: TC1 delay E-09/E-10/E-12; TC2 outage E-04/E-08/E-11; TC3 quality E-03/E-14/E-15; TC4 noise E-05/E-06/E-07; TC5 compound E-01; TC6 unconfirmed E-02/E-13.

6. What I would improve

- Desk follow-ups that also disambiguate shipments when several open ETAs match.
- Stronger grounding so ungrounded LLM rationales are never shown.
- Constraint solver for contested plant-day slots.
- Per-dealer notify templates.
- Streamed event bus instead of a batch of 15.

Model: qwen2.5:3b-instruct (Ollama, local). Stack: Python, networkx, Streamlit.

https://github.com/Darshil4154/hitachi-highway-control-tower